Replication of Study X by Sample & Sample (20xx, Psychological Science)

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2026-10-22

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Introduction

Proposal:

Zettersten & Saffran (2020) uses cross situational learning tasks to understand how learners gather new information for word learning. What do people choose to learn when presented with an uncertainty of new words? Experiment 1 asked whether or not adult participants, when given information to sample from, would choose information that reduces referential ambiguity. What sampling strategies are people using for word learning? This is the central

question that they later ask with a developmental population. They find that people choose the ambiguous word-object pair 62% of the time. The authors also ran a second version of Experiment 1 (S1), where the word-object ambiguity was graded instead across two between-subject conditions (Fully Ambiguous vs Partially Ambiguous). They replicated the findings from Experiment 1 and additionally also found that adults sampling preferences were sensitive to graded ambiguity.

This study aims to replicate the findings from the Supplementary Experiment 1 (S1) and also extend the study to see how sampling behavior changes over time by increasing the number of training-sampling rounds. We propose to first have participants complete S1 that is identical in structure and stimuli to replicate the original findings after which we would add the additional training-sampling rounds. This is a deviation from the original study, but allows for us to replicate the findings from S1 from the first round of sampling and testing trials.

Methods:

Stimuli: In S1, there were eight novel words and eight 'alien' images. The object-word association within each condition was randomized across participants. The independent variable in the experiment was the degree of ambiguity, manipulated as a between-subjects variable. There were two conditions: Fully Ambiguous in which the ambiguous object-word pairs always appeared together (the same as the main Experiment 1), and Partially Ambiguous in which the ambiguous object-word pairs appeared on only 4/6 of the trials. The key dependent variables were the sampling choice of objects that participants picked, reported as the proportion of ambiguous selections and test accuracy which determines whether participants are able to determine the correct object for each label. The experiment was developed in jsPsych (older version).

Design and Procedure: The Supplementary Experiment 1 consisted of four phases: Training, Sampling, and two Test Phases (whereas the main Experiment 1 only had one Test phase). The original structure of the S1 was Training-Test-Sampling-Test, while the proposed study's structure will be Training-Test-Sampling-Test-Sampling-Test-Sampling-Test-Sampling-Test.

Training Phase: participants are presented with two referents and two labels in a random order, occurring 6 times for a total of 24 times.

Sampling Phase: participants are presented with all 8 objects that appeared in randomized locations, where they are able to choose which items they would like to hear and see in the next set of cross situational learning trials. After this the, the object-label pair along with a second object chosen from random from the remaining objects in a cross-situational word learning trial with the same structure as in the training phase.

Testing Phase: participants finally completed an 8-AFC object-label recognition test without feedback.

Challenges The experiment was designed using some custom jsPsych plugins, which is where I believe I might run into some challenges. I aim to rebuild the entire experiment instead of using the plugins that are available on OSF.

Learning goals: My main learning goal throughout this quarter is to understand how to design, implement, and run better experiments. This includes understanding motivations behind a research question and how certain design choices can shape what a study examines. By recreating this experiment, it not only pushes me to have a thorough understanding of the methods but also allows me to creatively explore what the natural extensions to the study is and how we might be able to do it. Aside from the technical skills, I hope that this project allows me to leverage from existing code as a foundation to create my own. Beyond that, I would also like to understand the analyses in depth, learning not only how to do them in R, but what each analyses is measuring and what they mean.

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.